YAHIA ZAKARIA

AI Engineer · LLMs, Computer Vision & Data Analytics

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SUMMARY

AI engineer working remotely with a travel agency in Saudi Arabia. I build systems that turn Arabic conversations, documents and business data into numbers people can act on, and I test them on data they have never seen before trusting the result. Recent work: an Arabic data assistant that is 97.6% correct on a 200-question test, a document reader whose verified fields were 100% correct on 600 test documents, and a revenue analysis of 119,000 hotel bookings.

EXPERIENCE

Data Analyst / AI Researcher — OrenGreenberg · Remote · United Kingdom — **Jul 2026 – Present**

- Helping explore how AI is changing go-to-market and marketing work.

AI Engineer — Travel Gate KSA · Remote · Riyadh, Saudi Arabia — **Nov 2025 – Present**

- Built an LLM tool that analyses Arabic customer-support chats to score agent quality and surface service insights for managers.
- Built a pipeline that collects CRM conversations into a PostgreSQL database and enriches them with AI analysis (Python, FastAPI, n8n, Railway).
- Designed dashboards in Bitrix24 BI Builder (Apache Superset) tracking the sales pipeline, agent performance and marketing conversion across 9,000+ contacts.
- Delivered a demand analysis and 3-month forecast over 7,953 deals that supported a partnership decision with a major airline.
- Found and corrected misleading KPIs caused by data-quality issues, including an inflated conversion rate.

AI Engineer Trainee — National Telecommunication Institute (NTI) · Alexandria, Egypt — **Jul 2025 – Oct 2025**

- 540-hour "Hire Ready" programme in AI and data engineering.
- Built the charts layer of a team project that gathers competitor information from the web and turns it into market insights.

PROJECTS

Arabic Support Assistant *(Python, LLMs, RAG, sentence-transformers, DeepSeek)* — github.com/Yahia20/arabic-support-assistant
Answers travel-agency customers in Saudi dialect, formal Arabic or English from written policy; code checks every figure before an answer is sent and hands everything else to a person. 0 wrong answers of 163 sent and 68 of 68 hand-overs on a frozen 245-message test; the model costing 5x more sent 4 wrong answers, so the cheaper one ships ($0.17 per 1,000 chats).

KSA Document Reader *(Python, OpenCV, PP-OCR, ONNX Runtime)* — github.com/Yahia20/ksa-document-reader
Reads passports, IDs, visas, ZATCA invoices and boarding passes (10 document types) from photos and gives every field a status: verified, read or for review. 100% of verified fields correct on a frozen 600-document test; 99.86% on 3,315 real phone-video frames (MIDV-500).

Ask Your Data — Arabic data assistant *(Python, LLMs, SQL, PostgreSQL)* — github.com/Yahia20/ask-your-data
Managers ask business questions in Saudi dialect or formal Arabic and get the number, a chart and the query behind it. 31 KPIs defined once in a semantic layer; 97.6% correct on a 200-question test written before the code; asks a question back instead of guessing.

Hotel Booking & Cancellation Analytics *(SQL, DuckDB, dbt, Python, LightGBM)* — github.com/Yahia20/hotel-revenue-analytics
End-to-end analysis of 119,000+ bookings with 28 automated data checks. Cancellation model evaluated month by month on an unseen year (ROC AUC 0.85); a targeted deposit policy would have added an estimated €399k in eight months, where a deposit for everyone loses €199k.

Arabic Call Analyzer *(Python, Hugging Face, LangChain, Pydantic)* — github.com/Yahia20/arabic-call-analyzer
Call-centre QA for Arabic support calls: transcript, summary, call reason, whether the problem was solved, and an agent scorecard with a quote from the call backing every score.

SKILLS

LLMs & AI	LLMs, RAG, AI agents, LangChain, LangGraph, prompt engineering, LLM evaluation, Claude API
Deep learning & vision	PyTorch, Hugging Face Transformers, fine-tuning, OCR (PP-OCR), OpenCV, ONNX Runtime
Machine learning	scikit-learn, XGBoost, LightGBM, forecasting, time-based validation, leakage checks
Data & BI	SQL, PostgreSQL, DuckDB, dbt, pandas, Power BI, Apache Superset, Plotly
Engineering	Python, FastAPI, Docker, n8n, Railway, Git, GitHub

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services
HCCDA-AI — Huawei Certified Cloud Developer Associate in AI — Huawei, 2025
Machine Learning Specialization — Stanford Online · Coursera
Getting Started with Deep Learning — NVIDIA, 2025
Data Scientist Associate — DataCamp, 2025
Data Analyst Associate — DataCamp, 2026

EDUCATION

Bachelor's in Information Systems
Fayoum University — Faculty of Computers & AI · Graduated 2024

LANGUAGES

Arabic (Native) · English (Professional working)